

# ELU#

[https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.ELU.ht](https://pytorch.org/docs/stable/generated/torch.nn.modules.activation.ELU.html)

## Description:

Applies the Exponential Linear Unit (ELU) function, element-wise. Method described in the paper: Fast and Accurate Deep Network Learning by Exponential Linear Units (ELUs).  $\alpha$  (float) – the  $\alpha$  value for the ELU formulation. Default: 1.0 inplace (bool) – can optionally do the operation in-place. Default: False

## Function Signature

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```
class torch.nn.modules.activation.ELU(alpha=1.0,  
inplace=False)[source]#
```

Parameters

Shape:

```
extra_repr()[source]#
```

Return type

### Returns:

Tensor

## Code Examples

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```
>>> m = nn.ELU()  
>>> input = torch.randn(2)  
>>> output = m(input)
```

# Detailed Documentation

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## Documentation

Applies the Exponential Linear Unit (ELU) function, element-wise.

Method described in the paper: Fast and Accurate Deep Network Learning by Exponential Linear Units (ELUs).

alpha (float) – the  $\alpha$  value for the ELU formulation. Default: 1.0

inplace (bool) – can optionally do the operation in-place. Default: False

Input:  $(*)^{(*)}^{(*)}$ , where  $***$  means any number of dimensions.

Output:  $(*)^{(*)}^{(*)}$ , same shape as the input.

Return the extra representation of the module.

Runs the forward pass.